
title: "INFERNO Evaluation — A Hybrid Mathematical Framework for Morphogenesis and Regeneration" evaluator: Flow date: 2026-10-09 requested-by: Jan target: Cortés-Poza Y (2026) J Math Biol 93:43. DOI 10.1007/s00285-026-02459-2 (published, peer-reviewed; 40 pp.) author: Yuriria Cortés-Poza (IIMAS, UNAM, Mexico) codebook: INFERNO v1.0 (4 levels × 3 dimensions, genre-adapted: modeling-theory paper); WCI composite per meta/reference/wci.md basis: read-together — full primary text read in session; analytical results re-derived by hand; public code repo (YuririaCP/bioelectricity) cloned and reproduction run executed in the foundry status: complete (foundry reproduction results appended where available)

INFERNO EVALUATION — Cortés-Poza 2026 (JMB 93:43)

0. Genre identification

Modeling/theory paper in a peer-reviewed applied-math journal (Springer JMB). Single author. Constructs a hybrid multiscale dynamical framework (bistable bioelectric layer + adaptive gap junctions + 4-module synthetic GRN + slow tissue memory + wound wave + discrete morphology/occupancy) for planarian morphogenesis and regeneration. Three analytical results on reduced subsystems, seven open problems, three conjectures, four numerical experiments. No biological data anywhere — all "observations" are simulations of an explicitly uncalibrated model (unspecified time scale t^* , synthetic GRN, nominal parameters).

Caveat on this evaluation: the paper is self-declared as an architecture proposal ("This coarse-grained architecture suggests the following biological correspondences..." — future work). The WCI tier therefore measures use-readiness as a source of biological claims, not the quality of construction, which is good.

1. What the paper is and is not

- **Is:** a coherent, well-written, honest framework-construction paper; correct classical mathematics (re-verified); a genuinely integrative structure (dynamical systems + graph min-cut + Ising + Allen-Cahn + Hopfield memory + developmental biology); a public, runnable implementation.
- **Is not:** a validated model of regeneration. Nothing is calibrated to data; no signature planaria phenomenon (two-headed/octanol, 1/279th-fragment regeneration, depolarization reprogramming) is reproduced in-silico; the distinctive mechanism (slow memory ε) lacks a clean demonstration; numerics are 1D chains of 16-30 cells.

2. The INFERNO matrix (L1-L4 × D1 Data / D2 Formal / D3 Programmatic, 0-100)

Level	D1 Empirical/Data	D2 Formal/Analytical	D3 Programmatic
L1 Observation —	40 PARTIAL — no biological data; numerical observations on tiny 1D	45 PARTIAL — model-behavior observations; no data-driven inference	50 PARTIAL — observations motivate

Level	D1 Empirical/Data	D2 Formal/Analytical	D3 Programmatic
what does it expose?	systems (N=16-30), few seeds (3-15), illustrative sweeps		open problems/conjectures
L2 Inference — what formal machinery?	55 PARTIAL — standard RK4/projection numerics, clearly stated; code public (verified); dt inconsistency paper-vs-code (§5.1 says 0.01/0.02; repo validation runs 0.05)	65 PARTIAL/PASS — three analytical results, all correct (hand-verified) but thin: Prop 1 classical; Prop 2 near-tautological (inhibitor P has no spatial coupling) and explicitly does not extend to the full model; Theorem 1 standard Lyapunov/gradient. Energy-Ising-mincut-Allen-Cahn connections well-drawn, honestly flagged as analogies	70 PASS — the 4-layer hybrid architecture + timescale hierarchy is a genuinely reusable inferential structure; Hopfield analogy + Problem 7 give it programmatic reach
L3 Program evaluation — evaluates its own program?	60 PASS — positions vs Pietak-Levin 2016, Turing/Gierer-Meinhardt, Davidson GRN; identifies novelty (adaptive topology, slow memory, unprescribed target); but no quantitative head-to-head with any alternative on the same tasks	65 PARTIAL — ablations exist (R _i knockout, fixed-G, memory knockout) but are qualitative; the memory ablation's central contrast is not cleanly shown (both regimes switch at the tested amplitude)	65 PARTIAL — explicit about open problems; relations to experimental program (Beane 2011, Oviedo 2010, Reddien) well chosen
L4 Convergence — multiple traditions?	N/A	65 PARTIAL — genuine multi-tradition integration (dynamical systems, statistical physics, graph algorithms, PDE free-boundary, associative memory); the paper's real strength	60 PARTIAL/PASS — links attractor theory (Waddington/Kauffman) to bioelectric experimental program

3. Foundry reproduction of the numerical claims (see [run_reproduction_cortes.py](#) + [results/xenobot-p2i/cortes-poza/repro.log](#))

Reproduction run executed against the public implementation (github.com/YuririaCP/bioelectricity) with the paper's stated parameters. **CONFIRMED RESULTS (log preserved):**

- **[0] §5.1 energy-barrier formula error — CONFIRMED:** correct $\Delta U = 81a/64 = 1.5188 \rightarrow \delta = 1.743$; the paper's printed formula $(a/4)(V_0 - V_-)^2(V_+ - V_-)^2 = 6.0750 \rightarrow \delta = 3.486$. Factor-4 mismatch, reproduced in code.
- **[1] Nucleation threshold — CONFIRMED:** $p_{pat} = 0.000$ for $\delta \leq 0.7$ (paper: "essentially zero for $\delta \leq 0.7$ "); $\delta = 1.9 \rightarrow p_{pat} = 0.533$ (paper claims ≈ 0.53 at $\delta^* \approx 1.9-2.0$ — reproduces almost exactly); $\delta = 2.1 \rightarrow 0.800$; rise steep but noisy at 15 seeds ($\delta = 1.5 \rightarrow 0.133$, $\delta = 1.7 \rightarrow 0.400$, $\delta = 2.3 \rightarrow 0.400$). Energy-barrier prediction 1.74 sits inside the rising region.
- **[2] Regeneration robustness — CONFIRMED, cleaner than claimed:** lesion $\{0.1, 0.3, 0.5, 0.7\} \rightarrow p_{success} = 1.00$, fidelity = 1.00 across all (paper: fidelity ≥ 0.95).
- **[3] Basin boundary — CONFIRMED:** $f = 0.50$ ($f = 0.43 \rightarrow = -0.38$, collapses to V_- ; $f = 0.5 \rightarrow \approx 0$; $f = 0.57 \rightarrow = +0.19$), matching the claimed $f \approx 0.5$.
- **[4] Polarity reversal — CONFIRMS the ϵ -critique (C4/I-issue):** at $A = 2.5$, $T_{force} = 12$ BOTH regimes switch permanently — $\tau_{\epsilon} = 50$: polarity $+2.99 \rightarrow -2.61$; $\tau_{\epsilon} = 0.3$: $+2.64 \rightarrow -1.42$ (permanent, though the no-memory switch is less complete).

The discriminating role of slow memory is NOT isolated by the displayed experiment, as flagged.

- **[5] Lyapunov descent — CONFIRMED:** pure bioelectric $\Delta E = -5.78$ (<0), full model $\Delta E = -1.21$ (<0); effective dissipativity (Remark 8) holds in-reproduction.
- **[6] Switching phase diagram — CONFIRMED:** $A = [2.44, 2.44, 2.06, 2.06, 2.06, 1.67, 1.67, 1.67, 1.28]$ over $T_force = [2..20]$; A non-increasing in T_force (matches Problem 3/Conjecture 3); for $T_force \geq 6$, $A^* \approx 2.06$, within the paper's claimed 2.0–2.5 band. Full matrix: no switch below $A \approx 1.3$, all-switch above $A \approx 2.4$.

Additionally: **§5.1 vs Fig 2 caption seed-count contradiction** ("15 independent seeds per value of δ " vs "30 seeds per δ value") confirmed as an unresolved internal inconsistency.

4. PQS — Prediction Quality Score (modeling-paper weights)

Criterion	Weight	Score	Note
Pre-specified	0.30	50	Experiments read as pre-specified; several central claims post-hoc (ε role; "not prescribed")
Quantified	0.20	60	$\delta \approx 1.9$ – 2.0 , $f \approx 0.5$, $A^* \approx 2.0$ – 2.5 , fidelity ≥ 0.95 — all quantified
Falsifiable	0.20	45	As a biological model: not falsifiable yet (uncalibrated, no data confrontation); as a mathematical framework: checkable
Discriminating	0.20	40	No quantitative comparison against alternative models on the same tasks; no decisive test isolating ε or adaptive-G
Proportional	0.10	55	Claims $\approx$ evidence at the model level; several overreaches ("not prescribed externally", "consistent with the Lyapunov analysis")

$$\text{PQS} = 0.30 \cdot 50 + 0.20 \cdot 60 + 0.20 \cdot 45 + 0.20 \cdot 40 + 0.10 \cdot 55 = 50/100 \text{ (Moderate-Low)}$$

5. WCI composite (0–100, 6 dimensions, equal weights)

Dimension	Score	Rationale
Theoretical coherence	72	Architecture coherent; analytic results correct; minor tensions: δ^* formula error, no-Turing framing, $\tau_GRN = \tau_G = 5$ (no strict separation), "structural robustness" asserted for 40+ parameters
Empirical support	48	Zero biological data; all evidence is simulation of an uncalibrated model; claims about biology unsupported by data (framed as future work)
Replicability	75	Code public, seeds fixed, deterministic; reproduction run in progress in foundry — strong by construction
Independent uptake	10	Published Aug 2026; no citations yet; single-author; repo 0 stars (time-dependent)
Explanatory power	58	Coherent organizing framework for the bioelectric program; explains only in-silico behavior; ε -mechanism weakly demonstrated
Falsifiability	42	Not confrontable with data in current form (no calibration protocol, synthetic GRN, nominal parameters); analytic claims are checkable and mostly check out

WCI = (72+48+75+10+58+42)/6 = 50.8 → Tier 3 (borderline T2). Interpretation: as a construction to build on, this is a T2-quality framework; as a source of biological claims, it is not usable yet (T3). The gaps are structural but straightforward to close: calibration targets, a near-threshold memory-contrast experiment, 2D results, and the signature-phenomena simulations would move it decisively into T2. Unlike the Le preprint, the path forward here is code-and-experiments, not artifacts-and-accounting.

6. Net verdict

Correct, honest, well-integrated, reproducible framework paper whose scientific content is architectural rather than empirical. The mathematics verifies; the numerics are reproducible (foundry run pending); the claims about biology are scoped as proposals and should be read as such. The distinctive contributions — slow tissue memory and adaptive gap junctions as the mechanism of permanent reprogramming — are the least-demonstrated parts of the paper, and the "unprescribed morphology" framing is weaker than the protocol supports. Recommendation to a reader: use as a toolbox and hypothesis generator; do not cite as evidence about planaria biology.